Hyper-Relational Knowledge Enhanced Network for Hypertension Medication Recommendation

Ke Zhang, Zhichang Zhang, *Member, IEEE*, Wei Wang, Yali Liang, and Xia Wang

***Abstract*—Hypertension is a prevalent cardiovascular disease that requires timely and precise medication management. However, previous medication recommendation studies have largely relied on analyzing electronic health records (EHR), overlooking the specialized knowledge required for hypertension treatment. Moreover, the hypertension-related knowledge contained in existing general medical knowledge graphs is overly simplistic, and the binary relation representations they employ fail to accurately represent the complex treatment logic, thus falling short of meeting medication recommendation needs. To tackle these concerns, we present a novel hyper-relational knowledge-enhanced hypertension medication recommendation model (HKRec). HKRec incorporates both professional treatment knowledge and individual characteristics of patients to provide personalized medication treatment plans. Specifically, a hyper-relational knowledge graph designed for hypertension medication treatment is first constructed. Next, we design a knowledge-driven encoder to capture the representations of hyper-relational knowledge within the graph, and develop an EHR-driven encoder to extract patient-specific features from the EHRs. By integrating medical knowledge entities and patient information, a recurrent mechanism is introduced to model the development process of patients' hypertension conditions, thereby enabling more effective medication recommendations. Results from experiments on real-world MIMIC-III and MIMIC-IV datasets demonstrate that the HKRec model outperforms several competitive baseline methods. The approach enables physicians to create more accurate and personalized medication plans, leading to better management of hypertension and improved patient outcomes. Our code is publicly accessible at https://github.com/zk0814/HKRec.**

***Index Terms*—Electronic health records (EHR), hyper-relational knowledge, hypertension, medication recommendation, recurrent mechanism.**

Received 7 April 2024; revised 20 September 2024 and 26 October 2024; accepted 29 October 2024. Date of publication 15 November 2024; date of current version 2 June 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62163033, in part by the Talent Innovation and Entrepreneurship Project of Lanzhou, China under Grant 2021-RC-49, in part by the Natural Science Foundation of Gansu Province, China under Grant 22JR5RA145, Grant 21JR7RA781, Grant 21JR7RA116, and in part by the Major Research Project Incubation Program of Northwest Normal University, China under Grant NWNU-LKZD 2021-06. The funding organizations had no role in study design, data collection and analysis, decision to publish, or manuscript preparation. *(Corresponding author: Zhichang Zhang.)*

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Digital Object Identifier 10.1109/TCSS.2024.3489973

I. Introduction

HYPERTENSION is one of the most prevalent chronic conditions, serving as the primary risk factor for stroke, myocardial infarction, and cardiovascular mortality [1]. Therefore, precise and timely pharmacological management of hypertension is of paramount importance. In clinical practice, physicians need to assess not only the overall health of patients, but also the impact of comorbidities on medication decisions. In addition, the management of hypertension often involves multiple medications, increasing the risk of harmful drug–drug interactions (DDIs). Consequently, physicians must skillfully balance treatment effectiveness with the potential risks associated of medication interactions, relying on their expertise and clinical experience. However, the large number of hypertensive patients poses a significant challenge for physicians to provide individualized treatment.

With the rapid advancements in deep learning, a multitude of medication recommendation methods have emerged, offering reliable medication advice to physicians. These methods primarily focus on modeling electronic health records (EHR) [2], [3], utilizing patients' diagnoses, procedures, and medication records to capture their health conditions. By simulating medication recommendations based on physicians' prescribing practices, these methods assist in guiding appropriate medication decisions. Some studies have also integrated knowledge of medication molecular structures [4], [5] and other relevant factors to enhance the performance of medication recommendation systems. However, in the clinical practice scenario, physicians often consider clinical treatment guidelines, personal experience, and patient-specific circumstances when making medication decisions, as shown in Fig. 1. Therefore, it is crucial to integrate professional knowledge of hypertension management with information from EHR when constructing medication recommendation models. The integration ensures that the models accurately reflect the clinical reality and provide physicians with personalised medication advice based on scientific evidence.

In the domain of medication recommendation, the utilization of medical knowledge graphs to merge patients' clinical data with medical knowledge is a common strategy. However, established medical knowledge graphs such as unified

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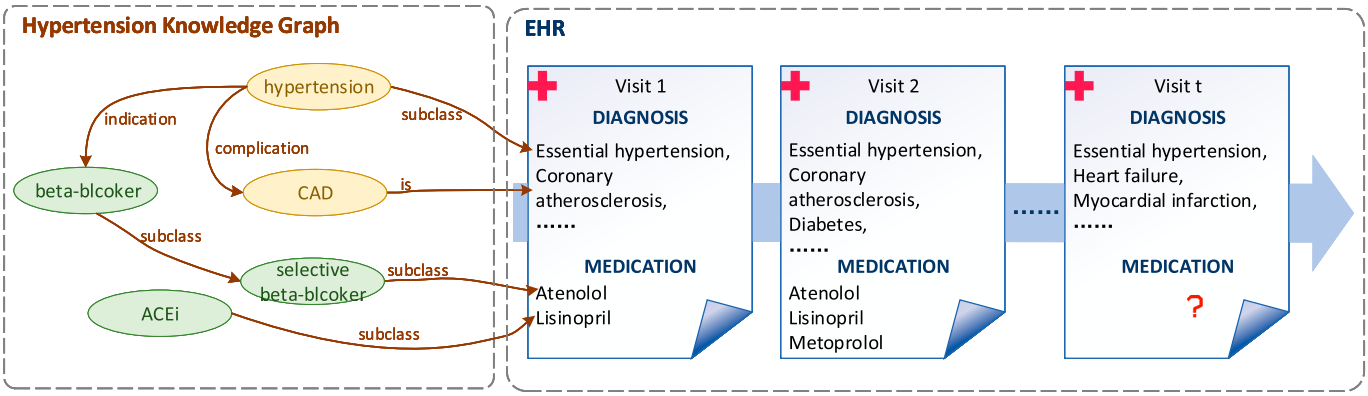

Fig. 1. Example of physicians prescribing medication. In clinical visits, physicians rely on a combination of clinical treatment guidelines and the patient's EHR to make appropriate medication decisions.

medical language system (UMLS) [6] and Chinese medical knowledge graph (CMeKG) [7] fall short in providing adequate support for hypertension medication recommendations. Existing medical knowledge graphs predominantly focus on fundamental medication classification and mechanisms of action, lacking the specific knowledge needed for hypertension treatment. But hypertension management encompasses multiple complex aspects such as medication selection, dosage adjustment, and management of complications, all of which require more detailed medical knowledge for guidance. Moreover, traditional knowledge graphs typically employ the subject-predicate-object (SPO) triplet structure to describe relationships between entities, which proves insufficient in capturing the complexity of hypertension treatment knowledge. For instance, consider the knowledge statement: *For hypertensive patients with diabetes, microalbuminuria, and symptoms of angina, beta-blockers and diuretics are preferred as the first-line antihypertensive medications* [8]. It is evident that this knowledge involves multiple entities and complex relationships. Apart from the explicit binary relationship of *preferred*, it also implies relationships such as *having a disease* and *having a symptom*. Obviously, the knowledge mentioned above cannot be adequately expressed using only three binary relationships. Although it is possible to combine several binary relationships, this results in a lack of independence in individual binary relationships, which can lead to ambiguity in knowledge representation. Therefore, to accurately represent the relationships between multiple entities, we propose incorporating of more advanced knowledge representation methods, such as hyper-relations.

As highlighted earlier, comprehensive hypertension treatment knowledge plays a pivotal role in medication recommendation. In this study, we introduce a novel hyper-relational knowledge enchanced network for hypertension medication Rrecommendation, known as HKRec. The model combines hypertension treatment knowledge with electronic health record data from both a knowledge-driven and EHR-driven perspective. We begin by exploring the mechanisms for representing and storing hypertension-related knowledge. Building upon this foundation, we construct a dedicated hyper-relational knowledge graph specifically focused on hypertension medications. Subsequently, we introduce a knowledge-driven encoder to comprehend the intricate representation of medical entities within the hyper-relational knowledge graph, as well as an EHR-based encoder to capture patient representations from EHR data. By effectively integrating medical entities from the knowledge graph and EHR data, we employ a recurrent mechanism to model the patient's historical health information and monitor the progression of hypertension condition. Comprehensive experiments on MIMIC-III [9] and MIMIC-IV [10] dataset confirm the effectiveness of the HKRec model, showing superior performance compared to existing methods.

To summarize, the key contributions of the article can be outlined as follows.

1) We present HKRec, an innovative hypertension medication recommendation model that effectively integrates expertise in hypertension treatment with patients' electronic medical record data. To our knowledge thus far, HKRec is the first study to introduce hyper-relational knowledge into medication recommendation.
2) We construct a hypertension-specific hyper-relational knowledge graph and devise a knowledge-driven encoder that captures both semantic and structural information related to hypertension treatment knowledge. Additionally, we incorporate a recurrent mechanism to proficiently capture the historical health information of patients.
3) The superiority of HKRec over existing approaches is supported by the experimental results obtained from MIMIC-III and MIMIC-IV datasets. Additionally, the conducted ablation study further substantiates the effectiveness of the hypertension hyper-relational knowledge graph we constructed.

II. Related Work

A. Medication Recommendation

In recent years, the widespread adoption of EHR has significantly enhanced the availability of patient visit information, which is crucial in the healthcare sector [11], [12], particularly

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in the domain of medication recommendation [13]. Traditional medication recommendation systems used content filtering, collaborative filtering, or hybrid filtering methods [14], but faced challenges such as data sparsity, cold start issues, and inadequate penalties [15]. To address these problems, researchers have turned to deep learning techniques to enable effective medication recommendation.

Deep learning approaches primarily achieve the objective through two medication recommendation methods: instance-based and longitudinal-based [16]. Instance-based models primarily utilize the patient's current medical information, including diagnoses, procedures, and laboratory test results, to generate recommended medications. For example, Zhang et al. [17] approached medication recommendation as a sequence decision problem and used a recurrent decoder to model label dependencies for recommending the most suitable medication. Likewise, Wang et al. [18] represented the patient's present health status as a heterogeneous graph and employed graph neural networks to recommend safe medications. However, instance-based models overlook the longitudinal medical history of patients and disregard disease progression or changes in treatment strategies. Consequently, these models may lead to inaccuracies in medication recommendations, such as prescribing the same treatment to a new patient with high blood pressure and a patient with chronic hypertension.

In contrast, longitudinal models place more emphasis on patient's medical history and medication records to provide appropriate medication recommendations. For example, Choi et al. [19] proposed the RETAIN model, which utilizes the attention mechanism to precisely identify historical visits that have a substantial impact on the treatment process, as well as key clinical variables. However, RETAIN overlooks potentially harmful interactions between medications. Shang et al. [16] modelled the patient's longitudinal records as queries and provided personalised safe medication recommendations based on a knowledge graph of medication interactions. Wang et al. [20] captured temporal information from medical records to gain insights into complex medication combinations. Additionally, An et al. [2] modelled temporal dependencies using an interactive long short-term memory network and captured hierarchical relationships between different sequences to simulate the decision-making process of physicians. Le et al. [21] extended the memory of recurrent neural networks using self-attention mechanism, enabling more accurate medication prediction based on patient history. Wu et al. [22] retrieved and mined the intricate relationships between a patient's historical diagnoses, medication recommendations, and current diagnosis. They also introduced a copy mechanism to determine whether to copy previous recommendations or generate new medications for each prediction. To further explore the relationship between medications and diseases, MoleRec [4] simulated the interactions between medication molecular structures and the impact of medication molecules on diseases, thereby improving the accuracy of medication recommendations while adhering to safety constraints. Despite significant advancements in existing research, these models are limited to utilizing EHR and overlook the importance of domain-specific medical knowledge. It is imperative to incorporate domain-specific medical knowledge to further enhance the accuracy and reliability of medication recommendations.

B. Hypertension Knowledge Graph

The medical knowledge graph is a structured representation and storage method for information in the medical domain [23]. It utilizes a graph format that includes medical entities and corresponding relationships. Entities represent various objects in the medical field, such as diseases, symptoms, and medications, while relationships illustrate the connections and interactions between these objects. By employing deep learning techniques, the entities and relationships within the knowledge graph can be embedded as vector representations, thereby enhancing the efficiency of reasoning medical information. The framework for knowledge graph embedding offers significant assistance for various subsequent tasks, including the development of medication recommendation systems.

However, there are challenges in current medical knowledge graphs. On one hand, general medical knowledge graphs lack the detailed information necessary to address diseases such as hypertension, failing to adequately reflect the complexity and nuances involved in hypertension management. On the other hand, current knowledge graphs typically represent knowledge using SPO triples, in which the head entity is the subject, the relationship is the predicate, and the tail entity is the object. While triples are powerful for modelling, they struggle to express the intricate multirelational of medical knowledge. To address the concern, some studies [24], [25] have introduced the concept of hypergraphs. Hypergraphs offer a solution by eliminating the binary constraint of relationships and representing knowledge as $H = (N, E)$, where N represents the set of nodes and E represents the set of edges, each of which can connect two or more nodes. However, the mixing of multiple entities within a relationship can lead to ambiguity in the specific meaning of the relationship. The introduction of hyper-relational graphs addresses the issue by representing knowledge as a combination of a main triple combined with multiple sets of key-value pairs [26], [27]. Specifically, hyper-relational graphs use the format $(h, r, t) + (k_1, v_1, k_2, v_2, \cdots)$ to represent knowledge that involves multiple entities, preserving the precise meaning of the relationship. Therefore, it is reasonable to implement hyper-relational graphs for medical knowledge representation.

III. Problem Formulation

A. Electronic Health Records (EHR)

The EHR encompasses visit information gathered from various patients. In the case of a particular patient, the EHR can be represented as a collection of records from multiple visits, denoted as $V = \left[V^1, V^2, \ldots, V^T\right]$, where V^i presents the ith visit and T indicates the total number of visits for patient. Specifically, the i-th visit of the patient can be denoted as $V^i = \left[V_d^i, V_p^i, V_m^i\right]$, where $V_d^i \in \{0,1\}^{|\mathcal{D}|}$, $V_p^i \in \{0,1\}^{|\mathcal{P}|}$, and $V_m^i \in \{0,1\}^{|\mathcal{M}|}$ are multi-hot vectors for diagnoses, procedures, and medications, respectively. Here $|*|$ is the total count of corresponding medical entity categories.

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TABLE I
NOTATIONS USED IN HKREC

Notation	Description
V	The longitudinal visit records for a single patient
V^i	The ith visit for a single patient
V_d^i, V_p^i, V_m^i	The vectors of diagnoses, procedures, and medications
G_e,G_d	The EHR or DDI graph
ε_*, A_*	The edge set and adjacency matrix of graph G_*
G_k	The hyper-relational knowledge graph
H_k, R_k, T_k	The head, relationship, and tail entities set in G_k
K_k, V_k	The key-value pairs set in G_k
$\hat{y}_t \in \{0,1\}^{\lvert\mathcal{M}\rvert}$	The multi-label prediction

B. Hyper-Relational Knowledge Graph

The hyper-relational knowledge graph covers concepts from various subdomains along with the intricate relations. The mathematical representation can be denoted as $G_k = \{H_k, R_k, T_k, K_k, V_k\}$, where H_k and T_k refer to the head and tail entities, respectively. The relationships between entities are represented by R_k, which includes diverse relationships (e.g., indication_to, is_subclass_of, contraindication_to). K_k and V_k represent the key-value pairs associated with the relationships.

C. EHR and DDI Graph

The EHR and DDI graphs can be represented as $G_e = \{M, \varepsilon_e\}$ and $G_d = \{M, \varepsilon_d\}$. Here ε_e encompasses comprehensive information on known medication combinations, while ε_d refers to the known DDI relationship between a pair of medications. To provide a elucidating explanation of the graph construction, adjacency matrices denoted as A_e, $A_d \in \mathbb{R}^{|M|\times|M|}$ are employed. In case of A_e, if the mth and nth medications coexist within a given visit, $A_e[m,n]$ is set to 1. Conversely, if a harmful interaction between the mth and nth medication is identified, $A_d[m,n] = 1$. G_e and G_d exhibit uniformity accross all patients.

D. Medication Recommendation Task

To recommend the appropriate antihypertension medication $\hat{y}_t \in \{0,1\}^{|\mathcal{M}|}$ for a patient, a medication recommendation model is trained using the patient's current diagnoses V_d^t and procedures V_p^t at time t, along with historical visit sequence $V = \left[V^1, V^2, \ldots, V^{T-1}\right]$, the constructed hypertension knowledge graph G_k, EHR graph G_e, and DDI graph G_d. For further elaboration, the notations used in this article are provided in Table I.

IV. METHODOLOGY

The structural design of the proposed HKRec model is detailed in this section, with the overall architecture depicted in Fig. 2. Initially, we create a hypertension hyper-relational knowledge graph (HKG) and develop a knowledge-driven encoder to generate accurate medical entity embeddings from the HKG. An EHR-driven encoder is then introduced to analyze clinical manifestations in EHR data. In order to consider medication coexistence and identify harmful interactions, we employ graph convolutional network [28]. By effectively combining HKG knowledge with EHR data and considering the chronic nature of hypertension, a medication recommendation module is implemented to address the longitudinal dependency of patients. Lastly, we discuss the loss function utilized for optimizing the medication recommendation network.

A. Hyper-Relation Knowledge Graph Construction

The knowledge graph is constructed using multiple established hypertension treatment guidelines [1], [8], [29], [30], [31], [32], [33], [34], [35], [36] as primary knowledge sources. To extract structured knowledge, we adopt a semi-automated approach due to the lack of computerization in these guidelines. Initially, we identify essential clinical knowledge and utilize entity recognition techniques [37] to extract various types of medical entities from medical guidelines. Subsequently, with the guidance of hypertension experts, we manually integrated medical entities to construct a knowledge system with triple or hyper-relational attributes. The process ensured an accurate and reliable representation of the knowledge.

Table II provides statistics on the quantity of entities within the knowledge graph and presents examples of these entities. Additionally, we define nine distinct relationships: *preferred*, *applicable*, *cautious*, and *forbidden* describe the associations between diseases and medication usage, while *comorbidity* characterizes the relationship between different diseases. *Indication* and *contraindication* are terminologies employed to indicate the suitability of medications for various diseases. *Numeric* refers to the patient's numerical indicators, while *subclass* signifies the hierarchical relationship between diseases or medications. Table III provides statistics for each type of relationship.

During the construction process, we represent the hierarchical knowledge of diseases and medications in the form of SPO triples. However, the complexity of knowledge regarding indications and contraindications of hypertension, which involves multiple entities, poses a challenge for accurate representation using SPO triples alone. To overcome this challenge, we utilize a hyper-relation graph to accurately represent the knowledge of indications and contraindications. Diseases are considered as head entities, while indications or contraindicated medications are treated as tail entities. Additional entities such as comorbidities, symptoms, and examinations are expressed using supplementary key-value pairs. For example, the knowledge *For hypertensive patients with diabetes and proteinuria, diuretic is the first-line antihypertensive medication* could be represented as (hypertension, *preferred*, diuretic) + (*comorbidity*, diabetes) + (*comorbidity*, proteinuria).

The knowledge graph we constructed stored 147 pieces of information related to hypertension treatment and 1682 pieces of information on treatment relationships between medications and diseases. Fig. 3 shows a portion of the hypertension knowledge graph constructed in this article, where arrows are used

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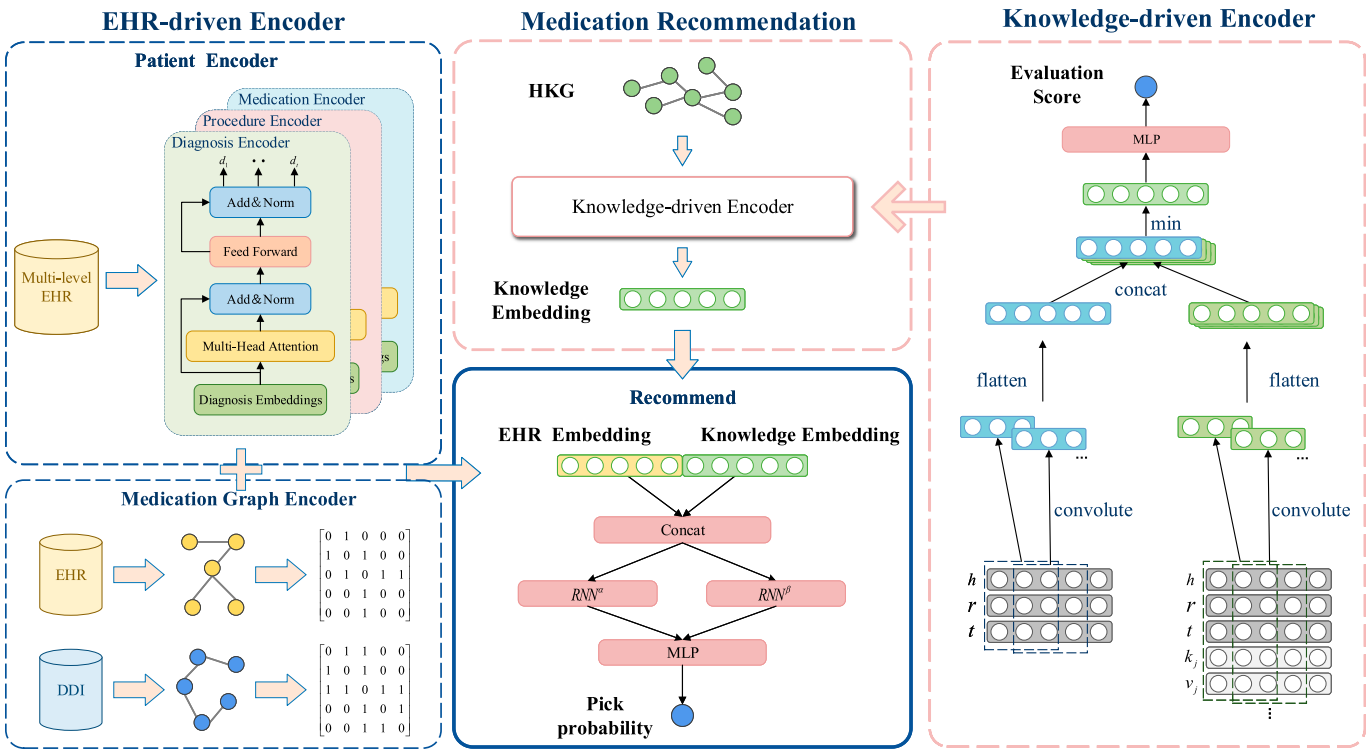

Fig. 2. Illustration of the HKRec framework. HKRec consists of three primary components: the EHR-driven encoder, the knowledge-driven encoder, and the medication recommendation module. The EHR-driven rncoder extracts the patient's historical visit information from the electronic health records (EHR), while the knowledge-driven encoder incorporates specific knowledge related to the treatment of hypertension. The outputs of these two encoders are combined, and the medication recommendation module generates the final medication predictions.

TABLE II
STATISTICS OF RELATIONSHIPS AND KEYS IN HKG

Entity Type	Number	Example
Disease	811	hypertension, diabetes
Numeric attribute	5	systolic blood pressure
Medication type	108	ACEi, ARB
Medication chemical	297	bisoprolol, hydrochlorothiazide

to label the SPO triple knowledge and the hyper-relational knowledge is visualised by dashed circles.

B. Knowledge-Driven Encoder

Given the constructed HKG, we use (h, r, t, k_j, v_j) to represent the hyper-relational fact, where $h \subseteq H_k$, $r \subseteq R_k$, $t \subseteq T_k$, $k_j \subseteq K_k$, $v_j \subseteq V_k$. The primary goal is to learn continuous representations of entities and relations, emphasizing the preservation of both structural and semantic information embedded within the graph.

For each hyper-relational fact, which consists of a base triple (h, r, t) and multiple associated key-value pairs $(k_j, v_j, j = 1, \ldots, n)$, we first extract the core information from the triple, as it embodies the primary structural details of the HKG. Thus, we utilize a convolutional neural network (CNN) [38] to model the relevance relationships within the triple, which represent the degree of semantic linkage and truthfulness in the embedding space. As shown in Fig. 2, the initial embedding vectors $h, r, t \in \mathbb{R}^k$ of the head, relation, and tail entities are concatenated to form a matrix $I \in \mathbb{R}^{3\times K}$. Subsequently, a 2-D convolution is applied using n_f filters of size 3×3 to capture the interdependencies between these entities. The output of n_f filters are then flattened as a vector $m \in \mathbb{R}^{1\times n_f(K-2)}$, representing the correlation of the base triple

$$m = \text{flatten}\left(\text{conv2}\left([h; r; t]\, W_m\right)\right). \tag{1}$$

To ensure the comprehensive representation of hyper-relational facts, we extend the modeling to capture the correlations between base triple and its associated key-value pairs. Similar to the process used for base triples, after obtaining the initial embeddings $h, r, t, k_j, v_j \in \mathbb{R}^k$ of base triple and a key-value pair, we concatenate them into a matrix $T \in \mathbb{R}^{5\times K}$. We then apply n_f filters of size 5×3 to model the correlations. The filter's first dimension, set to 5, is designed to correlations among the five elements h, r, t, k_j, v_j, while the second dimension, set to 3, aligns with the filter size used for base triplets, facilitating the integration of the resulting correlation features. The output of n_f filters are then flattened into a quintuple correlation vector. This process is repeated for each key-value pair (k_j, v_j) in the hyper-relational fact, where $j = 1, \ldots, n$, producing n distinct quintuple correlation vectors, denoted as $n_j \in \mathbb{R}^{1\times n_f(K-2)}$

$$n_j = \text{flatten}\left(\text{conv2}\left([h; r; t; k_j; v_j]\, W_n\right)\right) \tag{2}$$

where W_* is the learnable parameters of CNN filters.

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TABLE III
STATISTICS OF ENTITIES AND VALUES IN HKG

Knowledge Type	Preferred	Appliable	Cautious	Forbidden	Comorbidity	Indication	Contraindication	Numeric	Subclass
number	29	95	4	19	146	1478	204	14	717

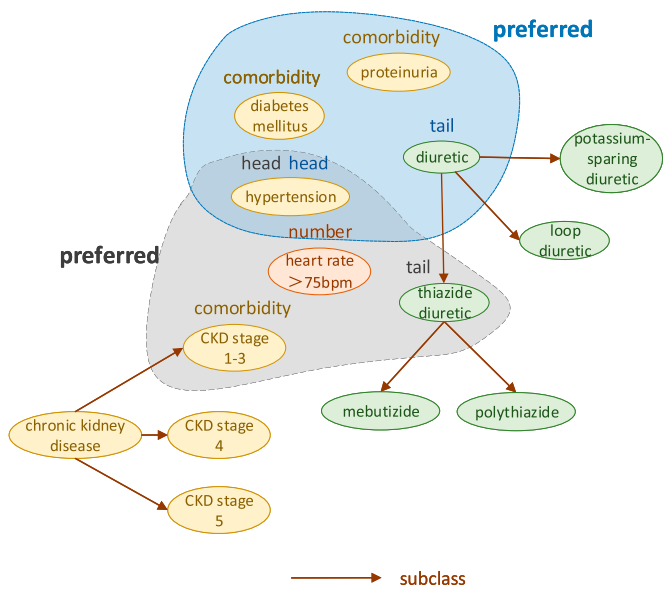

Fig. 3. Part of the constructed hypertension hyper-relational knowledge graph. The example contains two therapeutic knowledge in hyper-relational forms, three disease-level knowledge related to chronic kidney disease, and medication-level knowledge pertaining to diuretics.

Through the aforementioned steps, for each hyper-relational fact, we derive a triplet correlation vector from the base triplet (h, r, t) and n quintuple correlation vectors from the base triplet and its associated n key-value pairs. Subsequently, we concatenate the triplet correlation vector with the n quintuple correlation vectors to form a matrix of size $(n+1) \times n_f\,(K-2)$. We posit that in true hyper-relational facts, the correlations between the base triplet (h, r, t) and each key-value pair (k_j, v_j) should remain at a relatively high level. In other words, for the concatenated matrix, the minimum values across dimensions should not be excessively small. Therefore, we compute an overall correlation feature vector u by taking the minimum value along each column of the concatenated matrix, aiming to assess the authenticity of hyper-relational facts

$$u = \min\left([m; n_1; \cdots; n_j; \cdots]\right). \tag{3}$$

We employ a negative sampling strategy to train the module. For each triple or hyper-relational fact in our HKG, we generate negative facts by randomly replacing the tail entity of each positive fact. Subsequently, we optimize the module using a negative log-likelihood loss function, aiming for the module to assign higher scores to positive facts compared with negative ones:

$$L_{kg} = \sum_{F \in G_k} \log\left(1 + e^{-\alpha(F)}\right) + \log\left(1 + e^{\alpha\left(F'\right)}\right) \tag{4}$$

where G_k represents the input set containing all facts. $\alpha(F)$ denotes the predicted score generated by our knowledge-driven encoder for the positive fact F, while $\alpha(F')$ denotes the score for the negative fact F'.

C. EHR-Driven Encoder

1) Patient Encoder: To provide customised medication advice for hypertension patients using EHR data, we design a Transformer-based [39] encoder to effectively represent the patient's health status.

a) Self-attention: For a specific patient, given the query $Q \in \mathbb{R}^{q \times d}$, key $K \in \mathbb{R}^{k \times d}$ and value $V \in \mathbb{R}^{k \times d}$, we utilize the self-attention mechanism (SA) to model the entity-level correlation among all diagnoses within the same visit

$$\mathrm{SA}\left(Q, K, V\right) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d}}\right) V. \tag{5}$$

b) Multihead attention: Subsequently, we capture the interaction information from multiple perspectives

$$\mathrm{head}_i = \mathrm{SA}\left(QW_i^Q, KW_i^K, VW_i^V\right) \tag{6}$$

$$MH\left(Q, K, V\right) = \left[\mathrm{head}_1; \cdots; \mathrm{head}_h\right] W^H. \tag{7}$$

c) Position-wise feed-forward: To extract deeper-level features, we employ two linear projections and insert a ReLU activation function between them

$$H_d = \mathrm{LayerNorm}\left(V_d^t + MH\left(V_d^t, V_d^t, V_d^t\right)\right) \tag{8}$$

$$\mathrm{FFN}\left(H_d\right) = \mathrm{ReLU}\left(H_d W_1^F + b_1^F\right) W_2^F + b_2^F. \tag{9}$$

Finally, we obtain the encoded diagnosis representation as

$$e_{d_e}^t = \mathrm{Encoder}_d\left(V_d^t\right) = \mathrm{LayerNorm}\left(H_d + \mathrm{FFN}\left(H_d\right)\right). \tag{10}$$

Both the procedures representation $e_{p_e}^t$ and the medications representation $e_{m_e}^{t-1}$ are obtained through the same process

$$e_{p_e}^t = \mathrm{Encoder}_p\left(V_p^t\right) \tag{11}$$

$$e_{m_e}^t = \mathrm{Encoder}_m\left(V_m^{t-1}\right). \tag{12}$$

2) Medication Graph Encoder: In clinical practice, it is common for patients to use one or more medications at the same time. Physicians must carefully consider the potential harmful and beneficial interactions between medications when prescribing them. Therefore, modeling the coexisting relationships in EHR and harmful DDI relationships can improve the therapeutic effects of medications, as well as ensure the medication safety.

Inspired by COGNet [22], we employ the two-layer GCN to effectively encode two kinds of relationships derived from the

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EHR graph G_e and DDI graph G_d while mitigating high computational costs. Assuming $X^{l-1} \in \mathbb{R}^{M\times d}$ represents the input feature matrix of the lth layer, encompassing initial information of medication nodes. And the adjacency matrix $A \in \mathbb{R}^{M\times M}$ captures graph's topology, $\tilde{A} = A + I$, I is the identity matrix. The degree matrix is denoted as $D \in \mathbb{R}^{M\times d}$. The output of the lth layer is represented as

$$X^l = \text{GCN}\left(X^{l-1}, A\right) = \sigma\left(\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}X^{(l-1)}W^{(l-1)}\right). \tag{13}$$

Through (13), the first layer of GCN aggregates feature information of each medication node and its immediate neighbors. In the second layer of GCN, the output X^l from the first layer is used as the input feature matrix, thereby incorporating feature information from the two-hop neighborhood of each node. By employing the two-layer GCN, we derive the coexisting-aware medication representation G_e and DDI-aware medication representation G_d

$$\mathbf{G}_e = \text{GCN}\left(\text{ReLU}\left(\text{GCN}\left(X_m, A_e\right)W_e\right), A_e\right) \tag{14}$$

$$\mathbf{G}_d = \text{GCN}\left(\text{ReLU}\left(\text{GCN}\left(X_m, A_d\right)W_d\right), A_d\right) \tag{15}$$

where W_* represents the learnable parameter. Additionally, to effectively extract essential medication information from both the highly sparse graphs G_e and G_d, while also ensuring the stability of the model, we utilize ReLU as the activation function within the two-layer GCN.

To achieve a relation-aware representation, we combine medication representations of two relations: $\mathbf{G}_e$ and $\mathbf{G}_d$

$$G_M = \mathbf{G}_e - \lambda\mathbf{G}_d. \tag{16}$$

Finally, we obtain two representations $e^t_{m_m}$ and $e^t_{m_g}$ of the medication through the original embedding matrix X_m and the relation-aware matrix G_M. The two representations are fused to obtain the encoded medication representation

$$e^t_{m_e} = e^t_{m_m} + e^t_{m_g}. \tag{17}$$

D. Medication Recommendation Module

In this module, the first step is to collate the knowledge-driven representations into EHR-driven embeddings. Take the diagnosis information as an example

$$e^t_d = f\left(\left[e^t_{d_e}; e^t_{d_k}\right]\right) \tag{18}$$

where e^t_d represents the combined representation, which is obtained by applying a matching function $f(\cdot)$.

Then we use a dual-RNN to obtain the visit-level weight and medical entity-level weight, respectively. α_j represents the visit-level weight of each visit, and β_j is the attention weight of each medical entity within a visit

$$h^t_{d_\alpha} = \text{RNN}^\alpha_d\left(e^t_d, \ldots, e^1_d\right) \tag{19}$$

$$\alpha_j = \text{softmax}\left(W_\alpha h^j_{d_\alpha} + b_\alpha\right), j = 1, \ldots, t \tag{20}$$

$$h^t_{d_\beta} = \text{RNN}^\beta_d\left(e^t_d, \ldots, e^1_d\right) \tag{21}$$

$$\beta_j = \tanh\left(W_\beta h^j_{d_\beta} + b_\beta\right), j = 1, \ldots, t. \tag{22}$$

Algorithm 1: The Algorithm of HKRec.

Input: Hyper-relational knowledge graph G_k, EHR data, EHR graph G_e, and DDI graph G_d.

```
for each hyper-relational fact in G_k do
    Generate triplet correlation vector by Eq. (1);
    for each key-value pair in the hyper-relational fact do
        Generate quintuple correlation vector by Eq. (2);
    end for
    Compute overall correlation vector by Eq. (3);
end for
Optimize the entity representations by Eq. (4);
while not converged do
    Generate patient representations from EHR data using
    Transformer-based encoder by Eq. (5)-Eq. (12);
    Generate relation-aware medication representations
    from G_e and G_d using GCN by Eq. (13)-Eq. (17);
    Fuse the representations of diagnoses, procedures, and
    medications by Eq. (18);
    Compute attention weights for diagnoses, procedures,
    and medications by Eq. (19)-Eq. (23);
    Make medication predictions by Eq. (24);
    Optimize the overall model using the medication recom-
    mendation loss by Eq. (25);
end while
return Recommended anti-hypertension medications.
```

Output: Recommended medications.

The final representation of the patient's diagnosis at the tth visit is obtained by utilizing the generated attentions in the following manner:

$$d^t = \sum_{j=1}^{t} \alpha_j\beta_j \odot e^j_d. \tag{23}$$

After a series of similar processing steps, we eventually obtain the patient's ultimate procedure representation p^t and medication representation m^{t-1}. Finally, we use the patient representations for medication prediction:

$$\hat{y}^t = \sigma\left(\left[d^t; p^t; m^{t-1}\right]\right). \tag{24}$$

Based on previous research [19], [40], we formulate medication recommendation as a multi-label prediction problem. During the model training process, we employ the Adam optimizer [41] to optimize the binary cross-entropy loss function

$$L = -\sum_{t=1}^{T}\sum_{i=1}^{|\mathcal{M}|} y^t_i \log\hat{y}^t_i + \left(1 - y^t_i\right)\log\left(1 - \hat{y}^t_i\right). \tag{25}$$

V. Experiment

This section begins with a detailed description of the experimental setup. We then present the results of our experiments, demonstrating the effectiveness of our HKRec model.

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TABLE IV
STATISTICS OF PROCESSED DATASETS FOR ATC04 / ATC05 CODES

Item	MIMIC-III	MIMIC-IV
# patients	3,115 / 3,109	19 609 / 19 609
# visits	7,308 / 7,263	55 239 / 55 236
# diagnosis	1,966 / 1,965	2,000 / 2,000
# procedure	1,145 / 1,145	5,488 / 5,488
# medication	14 / 18	14 / 18
avg # of visits	2.3460 / 2.3361	2.8169 / 2.8169
avg # of diagnosis	10.8927 / 10.9262	9.3666 / 9.3666
avg # of procedure	4.0463 / 4.0609	2.5492 / 2.5492
avg # of medication	1.5508 / 1.4400	1.0804 / 1.1352

A. Dataset Description

To assess the performance of our model, we conduct experiments on two real EHR datasets: MIMIC-IIIv1.4[1] [9] and MIMIC-IVv2.2[2] [10] dataset. In particular, for the MIMIC-IV dataset, we focus on the electronic health records (EHR) data within the hospital scope, specifically the HOSP module. We extract relevant data pertaining to hypertensive patients based on their diagnosed ICD-9[3] codes. Subsequently, we perform data preprocessing, organizing the patients' diagnoses, procedures, and medication records. The dataset is then partitioned into three distinct sets: training, validation, and test, with a ratio of $(2/3):(1/6):(1/6)$, respectively.

Previous studies have primarily focused on providing recommendations for medication classes at the ATC04[4] level code. However, in real clinical settings, medication classes encompass a wide range of specific medications, making generalized recommendations less applicable. Our study not only anticipates medication classes at the ATC04 level (e.g., C02A, Antiadrenergic agents, centrally acting) but also extends to predicting medication types at the ATC05 level (e.g., C02AA, Rauwolfia alkaloids). The granular methodology enables a more precise and practical medication recommendation scheme. Statistical data can be found in Table IV, where only medication-related numbers are influenced by different medication classification criteria.

B. Baselines

We compare the performance of our approach against the representative methods.

1) *LR* refers to logistic regression with L2 regularization.
2) *ECC* [42] presents a classifier chain method that effectively models label correlations while maintaining reasonable computational complexity.
3) *RETAIN* [19] constructs a hierarchical attention model of capturing the relevance of different visits and variables.
4) *LEAP* [17] extracts meaningful representations during the current visit to generate medication sequences.
5) *DMNC* [21] focuses on the interaction of two asynchronous sequences by a memory-enhanced network.
6) *GAMENet* [16] is a graph-augmented memory network that incorporates DDI knowledge through a memory module and treats temporal patient records as queries.
7) *MICRON* [43] introduces a recurrent residual-based network, which is designed to incorporate changes in EHR as input and update the hidden medication set vector.
8) *SafeDrug* [40] is a DDI-controllable medication recommendation model that uses a message-passing neural network along with a bipartite learning method to model medication molecules.
9) *COGNet* [22] aims to predict each medication using a copy-or-generate module.

[1]https://physionet.org/content/mimiciii/1.4/
[2]https://physionet.org/content/mimiciv/2.2/
[3]https://en.wikipedia.org/wiki/List_of_ICD9_codes
[4]http://www.whocc.no/atc/structure and principles

C. Evaluation Metrics

To assess the performance of HKRec, we utilize several evaluation metrics: Jaccard Similarity Coefficient (Jaccard), precision-recall AUC (PRAUC), average F1 (F1), and DDI Rate (DDI). The definitions of these metrics are as follows.

1) *Jaccard*. For a specific patient, Jaccard is the ratio of intersection between the actual medication combination y^t and the predicted outcome $\hat{y}^t$ to their union

$$\text{Jaccard} = \frac{1}{T}\sum_{t=1}^{T}\frac{|y^t \cap \hat{y}^t|}{|y^t \cup \hat{y}^t|}. \tag{26}$$

2) *PRAUC*. For the tth visit of a given patient, PRAUC is a criterion to balance precision and recall

$$\Delta\text{Recall}(i)^t = \text{Recall}(i)^t - \text{Recall}(i-1)^t \tag{27}$$

$$\text{PRAUC}^t = \sum_{i=1}^{|M|}\text{Precision}(i)^t\Delta\text{Recall}(i)^t \tag{28}$$

$$\text{PRAUC} = \frac{1}{T}\sum_{t=1}^{T}\text{PRAUC}^t. \tag{29}$$

i represents the rank in the medication sequence. $\text{Precision}(i)^t$ is the precision at cut-off i within the ordered retrieval list, while $\Delta\text{Recall}(i)^t$ quantifies the change in recall upon retrieval of the i-th medication.

3) *F1*. For a certain patient, F1-score is the harmonic mean of precision and recall, which are defined as

$$\text{Precison}^t = \frac{|y^t \cap \hat{y}^t|}{|\hat{y}^t|}, \text{Recall}^t = \frac{|y^t \cap \hat{y}^t|}{|y^t|} \tag{30}$$

$$\text{F1}^t = \frac{2 \times \text{Precision}^t \times \text{Recall}^t}{\text{Precision}^t + \text{Recall}^t}. \tag{31}$$

We calculate the F1 score for for each visit and obtain the average F1 across all visits for the patient

$$\text{F1} = \frac{1}{T}\sum_{t=1}^{T}\text{F1}^t. \tag{32}$$

4) *DDI*. For DDI, the definition for a given patient is

$$\text{DDI} = \frac{1}{T}\sum_{t=1}^{T}\frac{\sum_{i=1}^{|\hat{y}^t|}\sum_{j=i+1}^{|\hat{y}^t|}1\left\{A_d\left[\hat{y}_i^t, \hat{y}_j^t\right]=1\right\}}{\sum_{i=1}^{|\hat{y}^t|}\sum_{j=i+1}^{|\hat{y}^j|}1}. \tag{33}$$

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TABLE V
PERFORMANCE COMPARISON ON DATASETS OF ATC04 / ATC05 CODES

(a) MIMIC-III

Model	Jaccard ↑	PRAUC ↑	F1 ↑	DDI ↓	Avg.# of Medications
LR	0.5203 / 0.4919	0.8117 / 0.7726	0.6347 / 0.5972	0.2951 / 0.3659	2.2480 / 2.0850
ECC	0.5438 / 0.5056	0.8278 / 0.7846	0.6593 / 0.6127	0.3012 / 0.3979	2.3070 / 2.0860
RETAIN	0.5540 / 0.5210	0.8143 / 0.7874	0.6783 / 0.6449	0.3921 / 0.5068	2.7191 / 2.6409
LEAP	0.5228 / 0.5082	0.6346 / 0.6162	0.6405 / 0.6217	0.2627 / 0.3648	**1.9388** / 1.9390
DMNC	0. 5342 / 0.5097	0.8225 / 0.7696	0.6463 / 0.6277	0.2365/ 0.2615	2.4580 / 2.3450
GAMENet	0.5497 / 0.5169	0.8239 / 0.7861	0.6710 / 0.6308	0.2366 / 0.4199	2.4569 / 2.3284
MICRON	0.5352 / 0.5146	0.8285 / 0.7884	0.6423 / 0.6145	0.2357 / 0.2658	1.9634 / 1.8059
SafeDrug	0.5474 / 0.5063	0.8252 / 0.7764	0.6484 / 0.6146	**0.2329 / 0.1975**	2.0000 / **1.7610**
COGNet	0.5325 / 0.5075	0.7972 / 0.7762	0.6448 / 0.6167	0.2902 / 0.3587	5.1720 / 4.6882
HKRec	**0.5918 / 0.5481**	**0.8422 / 0.8021**	**0.7101 / 0.6698**	0.2963 / 0.3890	2.6223 / 2.6292

(b) MIMIC-IV

Model	Jaccard ↑	PRAUC ↑	F1 ↑	DDI ↓	Avg.# of Medications
LR	0.4248 / 0.3518	0.7426 / 0.6873	0.5125 / 0.4315	0.2821 / 0.2734	1.5240 / 1.4390
ECC	0.4389 / 0.3779	0.7498 / 0.6923	0.5305 / 0.4624	0.3064 / 0.3047	1.5950 / 1.5370
RETAIN	0.5349 / 0.5045	0.7964 / 0.7641	0.6356 / 0.6008	0.3431 / 0.3197	2.0430 / 2.0479
LEAP	0.4186 / 0.3884	0.5047 / 0.4683	0.5326 / 0.4932	**0.0104 / 0.0506**	**1.2622 / 1.3184**
DMNC	0.5139/ 0.4787	0.7678 / 0.7277	0.5392 / 0.5094	0.2801 / 0.2792	1.9154 / 1.9396
GAMENet	0.5151 / 0.4738	0.7781 / 0.7326	0.6069 / 0.5662	0.2884 / 0.3050	1.9527 / 2.0695
MICRON	0.4297 / 0.3716	0.7543 / 0.7056	0.5151 / 0.4431	0.2866 / 0.2557	1.4890 / 1.5004
SafeDrug	0.5173 / 0.4767	0.7770 / 0.7313	0.6109 / 0.5648	0.3084 / 0.3255	1.9300 / 1.9745
COGNet	0.5087 / 0.4568	0.7448 / 0.7009	0.5948 / 0.5502	0.3261 / 0.3835	4.5811 / 4.8654
HKRec	**0.5707 / 0.5446**	**0.8159 / 0.7852**	**0.6660 / 0.6392**	0.3253 / 0.3262	2.2042 / 2.2333

In the above equation, A_d represents the adjacency matrix of the DDI graph G_d, while $\hat{y}_i^t$ denotes the tth recommended medication. The function $1\{\cdot\}$ returns 1 if the expression in $\{\cdot\}$ is true, and 0 otherwise.

D. Comparison Study

In this section, we comprehensively compare the performance and efficiency of HKRec with existing medication recommendation models.

1) Performance Comparison: The detailed results are presented in Table V, and ↑ signifies a preference for larger values, whereas ↓ signifies a preference for smaller values. Based on the analysis, we draw the following conclusions.

First, HKRec outperforms other methods in making medication recommendations. Specifically, the accuracy of recommendations is assessed using Jaccard, F1, and PRAUC scores, while safety is evaluated through DDI and Avg.# of Medications. Although HKRec excels in accuracy, its performance in safety is not optimal. Nevertheless, this indicates that HKRec strikes a good balance between accuracy and safety, making its medication recommendations useful in practice.

Second, LR and ECC perform poorly, likely due to their focus on the patient's current visit without considering long-term medical history. In contrast, models like RETAIN and GAMENet, which integrate historical medical records, show better performance, underscoring the importance of capturing patient history in medication recommendation models.

Third, in longitudinal model comparisons, RETAIN, GAMENet, and our HKRec model outperform other methods. These models utilize different recurrent mechanisms to model the progression of patient health and medication records. On the other hand, the MICRON model performs poorly due to its focus on modeling medication changes, but it struggles to capture long-term medication history. Additionally, LEAP tends to generate fewer drug combinations, resulting in optimal DDI outcomes. SafeDrug, on the other hand, prioritizes drug safety, which leads to significant shortcomings in other evaluation metrics.

Lastly, as medication codes are refined from ATC04 to ATC05, the complexity of the medication recommendation task increases, leading to a performance decline in all models. However, HKRec experiences the smallest decline in the top three key evaluation metrics, likely due to the detailed medication information included in the hypertension knowledge graph we constructed.

2) Efficiency Comparison: For leading deep learning-based algorithms, model efficiency is crucial. We evaluate the efficiency of the models, including the number of parameters, the average training time per epoch, and the number of epochs required for convergence. Additionally, we measure the time for a single round of inference on approximately 2,600 recommendation samples in the test set. Table VI presents the efficiency comparison of all models.

We can observe that models with higher time costs often use complex network architectures, particularly evident in the use of RNN and GCN [44]. Despite the larger number of parameters and higher time consumption, our model shows the fastest convergence speed during training. This suggests that under equivalent training conditions, our model can reach the desired level of performance more quickly, thereby reducing the overall training time. Furthermore, in the inference phase, our model

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TABLE VI
EFFICIENCY COMPARISON ON MIMIC-IV OF ATC04 CODES

Model	Parameter	Convergence	Training	Inference
RETAIN	535 247	49 Epoch	59.82s / Epoch	143.91s
LEAP	167 441	17 Epoch	245.77s / Epoch	684.17s
DMNC	765 352	31 Epoch	464.82s / Epoch	502.90s
GAMENet	681 423	22 Epoch	209.91s / Epoch	283.98s
SafeDrug	543 49	49 Epoch	209.91s / Epoch	262.22s
MICRON	507 726	35 Epoch	216.50s / Epoch	477.79s
COGNet	1 506 989	28 Epoch	743.72s / Epoch	521.59s
HKRec	1 494 291	15 Epoch	737.90s / Epoch	605.06s

offers superior accuracy and robustness compared to baseline models, which is crucial for medication decision support in real-world application scenarios.

To thoroughly analyze the efficiency issues, we then decompose the model and evaluate its complexity: 1) Knowledge-driven encoder module: This module leverages CNN to learn the representations of various medical entities within hyper-relational knowledge. The time complexity for learning triplets is $O\left(n_f \cdot \text{dim}\right)$, while for quintuplets, it is $O\left(n_k \cdot n_f \cdot \text{dim}\right)$, where n_f is the number of filters used in CNN, n_k represents the number of key-value pairs in a hyper-relational fact, and dim is the feature dimension. 2) EHR-driven encoder module: This stage employs Transformer-based mechanism and GCN for computation. The time complexity is influenced by the length of the input sequence n and the model dimension dim. In the Transformer encoder, the multihead self-attention mechanism has a time complexity of $O\left(h \cdot n^2 \cdot \text{dim}\right)$, where h is the number of attention heads, while the linear layers contribute $O\left(n \cdot \text{dim}^2\right)$. For the GCN, the time complexity is $O\left(n^2 \cdot \text{dim}\right)$. 3) Medication recommendation module: This module combines the embeddings of knowledge-driven encoder and EHR-driven encoder. The embedding concatenation incurs a time cost of $O\left(\text{dim}\right)$. The time complexity for the RNNs is $O\left(T \cdot \text{dim}^2\right)$, where T represents the number of time steps for each RNN unit. The final linear layer has a time complexity of $O\left(\text{dim}^2\right)$. Thus, the total complexity of the entire model is:

$$O\left(\left(n+T+1\right)\text{dim}^2 + \left(\left(1+n_k\right)n_f + \left(1+h\right)n^2 + 1\right)\text{dim}\right). \tag{34}$$

E. Ablation Study

To further validate the effectiveness of each module in the proposed HKRec model, we conduct the ablation studies. Our HKRec model is compared with the following variants.

1) HKRec $(w/o\ >$ KG): We exclude the information from the constructed Hypertension Hyper-relational Knowledge Graph. Therefore, the model's input only consisted of EHR and DDI information. The results presented in Table VII demonstrate that the original HKRec model outperforms this variant in providing more accurate medication recommendations, which can be attributed to the influential role of hypertension treatment knowledge in guiding the recommendation process.
2) HKRec $(w/o\ >$ KDE): We present an embedding approach tailored for hyper-relational knowledge graphs and compare it with the classical TransE [45] method for knowledge graph embedding. Our results show a notable reduction in medication recommendations across multiple evaluation metrics, suggesting that the knowledge-driven encoder module effectively captures intricate treatment knowledge pertaining to hypertension.
3) HKRec $(w/o\ >$ RNN): To validate the superiority of the longitudinal approach, we depart from using the RNN to capture the temporal dependency of each visit. Instead, we treat each visit as an independent training sample. In comparison to our longitudinal model, the variant exhibits poorer performance.
4) HKRec $(w/o\ > G)$: We remove the medication graph encoder module from the model to experiment. The results presented in Table VII show a noticeable decline in performance, especially in terms of safety. This emphasizes the importance of modeling the coexisting relationships of medications in EHR and DDI to improve the safety and effectiveness of medication recommendations.

F. Parameter Sensitivity

1) Influence of Varying Filter Numbers: In this section, we investigate the influence of the number of filters, represented as n_f, employed in the CNN architecture of the knowledge-driven encoder. Taking the ATC04 codes preprocessed MIMIC-IV data as an example. We initially tune the number of filters n_f with meal flavors ranging from 10 to 400, while keeping the embedding dimension K fixed at 64. Fig. 4 shows the effect of varying values of n_f on the performance of the medication recommendation task. Upon meticulous analysis of the experimental findings, it is apparent that the performance generally exhibits improvement with an increasing number of filters n_f. However, the performance stabilizes when $n_f \geq 300$. Therefore, to ensure optimal performance of the model, we adopt a fixed number of filters n_f, set at 300, and all subsequent experiments carried out within this research endeavour.

2) Influence of Varying GCN Layers: To ascertain the optimal number of layers for the graph convolutional network (GCN) employed in the medication graph encoder, experiments are conducted with varying settings using the MIMIC-III dataset with ATC04 level medication code, from one to four layers. As illustrated in Fig. 5, the model demonstrates optimal performance in the two-layer configuration. However, as the number of layers increases, performance declines due to over-smoothing. While the performance of three layers is comparable to that of two layers, adding further layers increases the complexity of the model. Consequently, we set the number of layers as 2 in this study.

3) Influence of Varying Visit Numbers: In the study of the medication recommendation module, we aim to comprehensively evaluating the capability of the Dual-RNN attention mechanism employed by HKRec in simulating patients' historical health records. To gain an in-depth understanding of the effectiveness of this mechanism, we specifically focus on how the recommendation results are affected by variations in the number of patient visits. Specifically, we examine the first five visits and the complete visit history of each patient

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TABLE VII
ABLATION STUDY OF HKREC ON DATASETS OF ATC04 / ATC05 CODES

(a) MIMIC-III

Model	Jaccard ↑	PRAUC ↑	F1 ↑	DDI ↓	Avg.# of Medications
HKRec $(w/o\,KG)$	0.5849 / 0.5415	0.8398 / 0.7983	0.7056 / 0.6612	0.3258 / 0.4015	2.6360 / 2.6630
HKRec $(w/o\,KDE)$	0.5832 / 0.5420	0.8350 / 0.8006	0.7011 / 0.6654	0.3218 / 0.3910	2.5543 / 2.7261
HKRec $(w/o\,RNN)$	0.5833 / 0.5413	0.8348 / 0.7998	0.7001 / 0.6646	0.3104 / 0.4143	2.5590 / 2.6477
HKRec $(w/o\,G)$	0.5836 / 0.5461	0.8323 / 0.8014	0.7012 / 0.6663	0.3142 / 0.4045	**2.5411** / 2.6809
HKRec	**0.5918 / 0.5481**	**0.8422 / 0.8021**	**0.7101 / 0.6698**	**0.2963 / 0.3890**	2.6223 / **2.6292**

(b) MIMIC-IV

Model	Jaccard ↑	PRAUC ↑	F1 ↑	DDI ↓	Avg.# of Medications
HKRec $(w/o\,KG)$	0.5595 / 0.5418	0.8109 / 0.7855	0.6552 / 0.6471	**0.3115 / 0.3189**	2.0710 / 2.2530
HKRec $(w/o\,KDE)$	0.5518 / 0.5384	0.8080 / 0.7812	0.6515 / 0.6402	0.3321 / 0.3338	**2.0519 / 2.2064**
HKRec $(w/o\,RNN)$	0.5623 / 0.5347	0.8086 / 0.7792	0.6557 / 0.6320	0.3250 / 0.3272	2.1789 / 2.1157
HKRec $(w/o\,G)$	0.5681 / 0.5321	0.8087 / 0.7780	0.6638 / 0.6356	0.3334 / 0.3353	2.2302 / 2.2608
HKRec	**0.5707 / 0.5446**	**0.8159 / 0.7852**	**0.6660 / 0.6392**	0.3253 / 0.3262	2.2042 / 2.2333

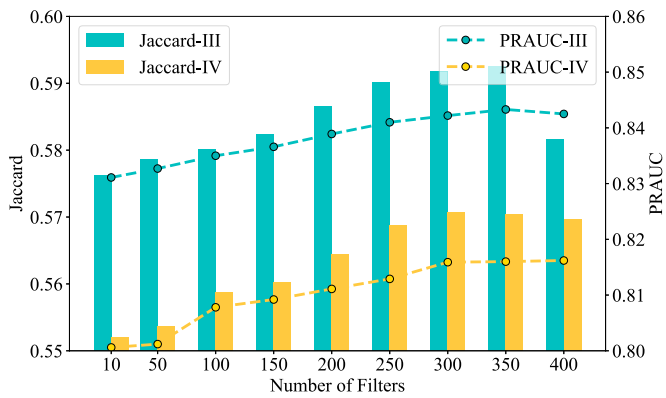

Fig. 4. Performance of HKRec with varying numbers of filters n_f.

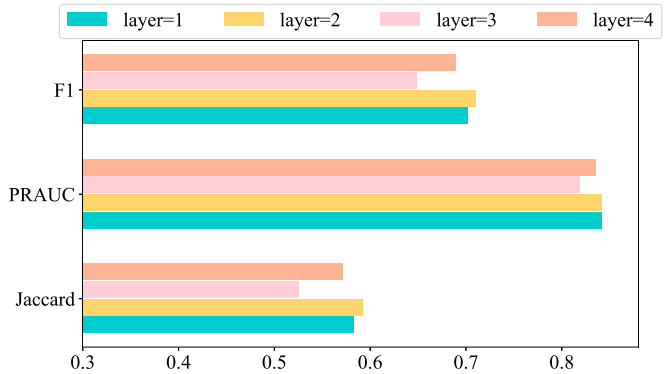

Fig. 5. Performance of HKRec with varying layers of GCN.

using the preprocessed data of ATC04 codes, as the majority of patients in MIMIC-III and MIMIC-IV datasets had fewer than five visits. Moreover, to ensure a comprehensive evaluation of HKRec's performance, two well-established baseline models, namely GAMENet and COGNet, were introduce for comparative analysis. These models are selected based on their established effectiveness in medication recommendation tasks.

The experimental results are illustrated in Fig. 6 demonstrate that as the number of visits increases, the performance of all three models generally exhibits an upward trend. Notably, HKRec is capable of rapidly capturing the initial growth in visit frequency during the first two visits, achieving significant performance enhancement. In contrast, GAMENet and COGNet require the accumulation of sufficient historical information before showing noticeable performance improvements.

Furthermore, both HKRec and GAMENet employ recurrent mechanisms to generate attention weights, resulting in relatively stable performance. HKRec leverages a dual-level recurrent mechanism to generate attention weights, selectively incorporating valuable historical visit information and clinical variables, thereby enhancing the accuracy of medication recommendations. In comparison, GAMENet relies solely on a visit-level recurrent mechanism to generate patient representations, resulting in inferior performance compared to HKRec. On the other hand, COGNet updates existing medication combinations by generating new sequences. While its replication or generation mechanism may offer novel medication suggestions, it also increases the risk of accumulating erroneous information, causing significant performance fluctuations during the initial visits. Overall, the Dual-RNN attention mechanism employed by HKRec effectively improves model performance.

G. Case Study

1) Overall Performance: To demonstrate the effectiveness of our constructed hypertension knowledge graph and the HKRec model, we randomly choose a patient from the test dataset. Due to space limitations, we use ICD-9 codes for diagnoses and ATC04 level codes to represent medications. In addition, the term "Ground Truth' pertains to the true prescriptions, "unseen" indicates medications that are predicted by the model but did not exist, and "missed" refers to medications that are prescribed but not predicted by the model.

As depicted in Table VIII, the patient's first visit revealed diagnoses of hypertension, coronary atherosclerotic heart disease, and myocardial infarction. The same diagnoses persisted at the

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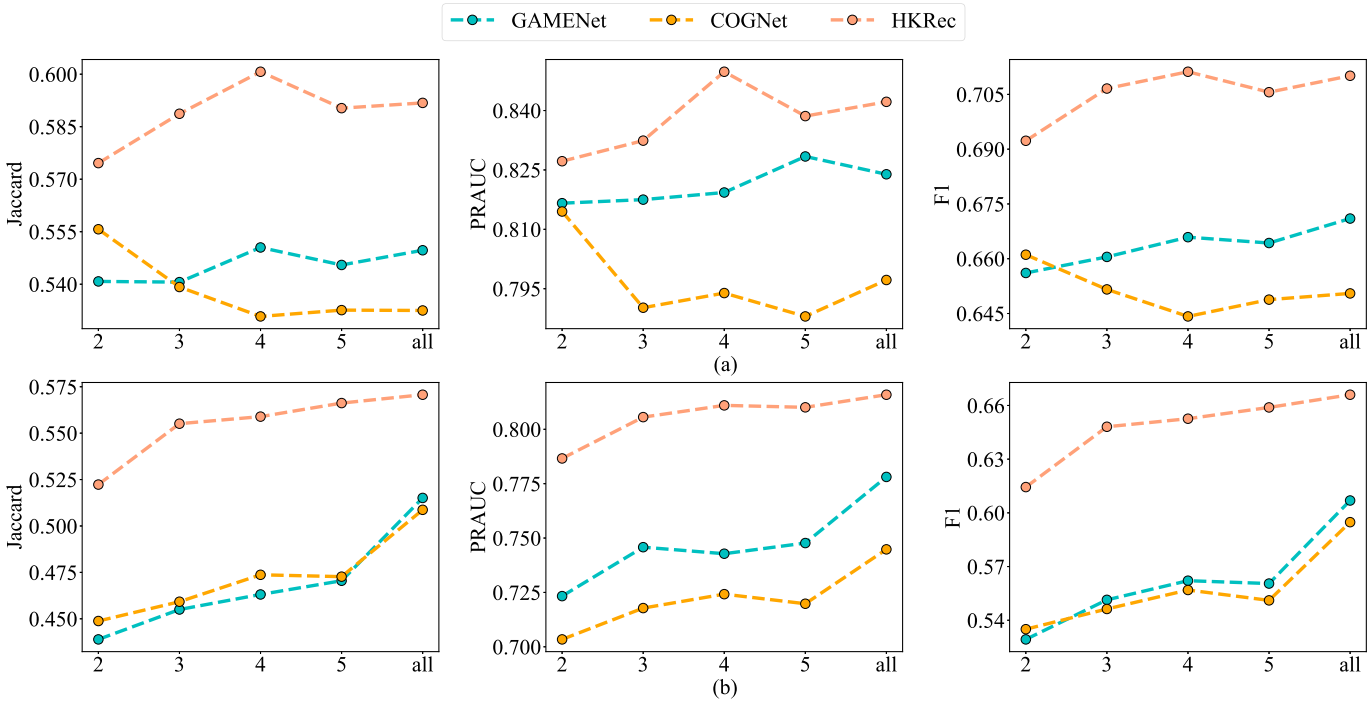

Fig. 6. Performance of different models under various number of visits. (a) MIMIC-III. (b) MIMIC-IV.

TABLE VIII
CASE STUDY OF HKREC ON MIMIC-IV

Diagnoses	Method	Recommended Medication
visit1: 4019,41400, 60000,53081, 412	Ground Truth	3 (C08C,C07A,C02C)
	GAMENet	1 correct(C07A) 2 missed(C08C,C02C) 0 unseen
	COGNet	1 correct(C08C,C02C) 1 missed(C07A) 0 unseen
	HKRec	3 correct(C08C,C07A,C02C) 0 missed 0 unseen
visit2: 1888,45829, 27652,4019, 41400,53081, 412,V173	Ground Truth	4 (C07A,C03C,C08C,C02C)
	GAMENet	2 correct(C07A,C02C) 2 missed(C03C,C08C) 0 unseen
	COGNet	3 correct(C07A,C08C,C02C) 1 missed(C03C) 1 unseen(C02A)
	HKRec	3 correct(C07A,C08C,C02C) 1 missed(C03C) 0 unseen

second visit, accompanied by additional diagnoses of hereditary psychiatric disorders, peptic ulcers, and hyperlipidemia. The table showcases the medications prescribed by physicians and various recommend methods. Overall, the HKRec model delivers the best performance, accurately recommending the first medication that aligns perfectly with the physician's prescription, with only one medication missing at the second visit. Conversely, GAMENet, which relies solely on information from the first two visits, falls short in recommending an adequate number of medications due to insufficient historical data. And COGNet makes an attempt to suggest additional medications but introduces an unseen error.

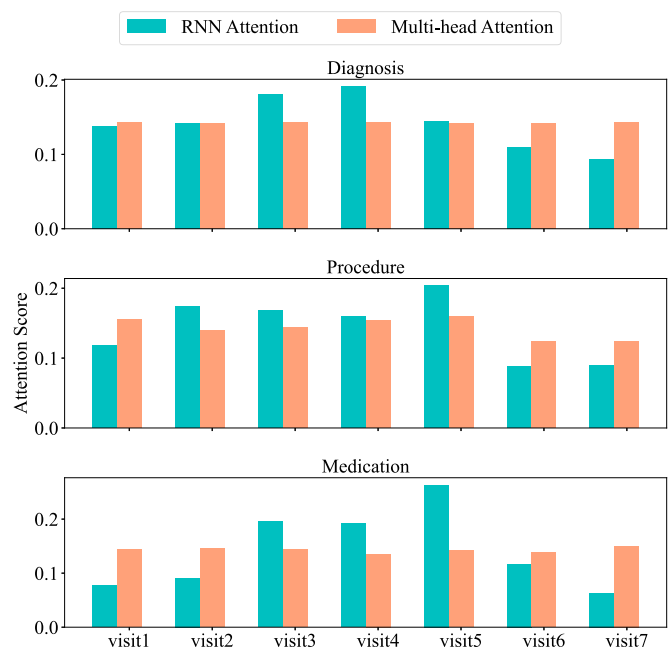

Fig. 7. Visualization of attention weights across historical visits.

Notably, at the second visit, COGNet recommends C02A (specifically reserpine, C02AA02) for the newly diagnosed hereditary psychiatric disorder. However, rifampicin, which is known to increase gastric acid secretion, is unsuitable for patients with peptic ulcers. Guided by hypertension treatment expertise, HKRec adeptly balances safety and efficacy considerations. Consequently, it accurately decides to exclude C02A, aligning perfectly with the physician's actual decision.

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2) Visualization of Attention Weights Across Historical Visits: To provide doctors with an intuitive explanation, we select a patient with eight visits and conduct a visual analysis of the recommendation. This analysis specifically showcase the distribution of attention weights across historical visits generated by the Dual-RNN attention mechanism in the Medication Recommendation Module. Additionally, we explore the multi-head attention as an alternative attention mechanism. As illustrated in Fig. 7, the Dual-RNN attention can generate sparse attention weights, assigning lower attention to less significant visits, such as visit7. In comparison, the attention weights produced by multihead attention exhibit less discrimination. During clinical decision-making, the HKRec model intelligently identifies which visits should be prioritized and which can be disregarded, enhancing the model's interpretability.

VI. Conclusion

The study introduces a hyper-relational knowledge enhanced network called HKRec for hypertension medication recommendation. The innovative approach represents the pioneering application of hyper-relational knowledge in the field of medication recommendation. Through experiments on the MIMIC-III and MIMIC-IV datasets, HKRec showcases a notable enhancement in hypertension medication recommendation, offering more precise and efficient medication guidance for hypertension treatment.

In the future, we aim to optimize HKRec by reducing computational complexity and execution time while maintaining high performance. We are also committed to expanding the scope of the hyper-relational knowledge graph. By integrating a wider range of medical literature, clinical trial data, and real-time patient feedback, we aim to build a comprehensive knowledge graph to provide physicians with more reliable and timely practice and policy recommendations, thereby improving the quality and effectiveness of healthcare services.

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